

Heuristic Autonomous Lunar Operation - Docking

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Autonomous rendezvous and docking in cislunar space requires robust onboard guidance that operates during periods of ground-based communication blackout. Current systems rely on multiple rendezvous sensors to estimate relative state, and this dependence introduces a critical vulnerability: loss of sensor data results in loss of guidance capability. This work introduces HALO-D (Heuristic Autonomous Lunar Operational Docking), a family of Decision Transformer-based guidance policies designed to perform closed-loop docking using onboard observations. Three model configurations are evaluated: a telemetry-only model, a vision-only model using RGB and depth image sequences, and a fused vision-telemetry model. All models are trained via behavioral cloning of a Linear Quadratic Regulator (LQR) expert within the SpaceTeamsPRO high-fidelity simulation environment and conditioned on a return-to-go (RTG) parameter that targets fuel-efficient trajectories. Performance is assessed by docking success rate, fuel consumption, contact speed, and command-tracking accuracy across closed-loop trajectories drawn from a randomized initial-condition distribution. All three models achieve at least 98% docking success while satisfying the soft-contact constraints. The vision-only model achieves performance statistically indistinguishable from the fused configuration, with mean fuel consumption of approximately 96.3 kg and docking duration of 72.1 s. These results demonstrate that a temporal sequence of depth and RGB images alone provides a sufficient state representation for closed-loop docking guidance, suggesting that a single onboard camera coupled with a learned guidance policy can replace traditional rendezvous sensor suites without sacrificing performance and enable lower-mass autonomous docking systems for future cislunar missions.

Nomenclature

$\mathbf{p}$	=	relative position vector, m
$\mathbf{v}$	=	relative velocity vector, m/s
$\mathbf{q}$	=	attitude quaternion (x, y, z, w), –
$\boldsymbol{\omega}$	=	body angular velocity vector, rad/s
u_τ	=	normalized six-dimensional action vector at step τ , –
T	=	final timestep of a trajectory, –
t	=	current timestep, –
C	=	context window length, frames
H	=	planning horizon length, steps
d_{model}	=	transformer hidden dimension, –
RTG_t	=	return-to-go scalar at timestep t , –
W	=	SVD sliding-window width, steps
N_f	=	SVD per-step feature count, –
K	=	number of retained SVD components, –
V_K	=	matrix of top- K right singular vectors, –
$\tilde{w}$	=	normalized window vector, –
$\varepsilon(w)$	=	per-window L2 reconstruction error (anomaly score), –
τ	=	SVD anomaly detection threshold, –
THC	=	translational hand controller (translational thrust command)
RHC	=	rotational hand controller (rotational moment command)
LQR	=	Linear Quadratic Regulator
ViT	=	Vision Transformer
MAE	=	Mean Absolute Error

I. Introduction

The Artemis program has renewed the operational demand for autonomous rendezvous and docking in cislunar space. Unlike low Earth orbit operations, cislunar trajectories routinely encounter extended communication blackouts during which ground-based intervention is infeasible, placing the full burden of the docking maneuver on the onboard guidance, navigation, and control system. The Orion vehicle and its peers in the cislunar architecture meet this requirement with classical guidance laws driven by dedicated rendezvous sensor suites such as LiDAR rangefinders, optical navigation

cameras, and inertial measurement units, all fused to recover relative position, velocity, and attitude. These pipelines are flight proven but inherit a structural weakness: any failure of the rendezvous sensor suite leaves the vehicle without a recovery path, since the classical controllers cannot operate without their state estimates.

This work introduces HALO-D (Heuristic Autonomous Lunar Operational Docking), a family of vision-based Decision Transformers trained to dock the Orion spacecraft from a temporal sequence of onboard observations. Three input configurations are evaluated: a telemetry-only model that consumes only the onboard 13-dimensional state, a vision-only model that consumes only RGB and depth image sequences, and a fused model that combines both. The vision-only configuration is the central research target. If a temporal sequence of RGB and depth images alone encodes range, lateral offset, approach rate, and attitude with sufficient fidelity for closed-loop guidance, then autonomous docking becomes possible on any spacecraft equipped with a camera, eliminating the dedicated rendezvous sensor suite as a single point of failure and reducing vehicle mass, cost, and integration complexity.

The three models are trained by behavioral cloning of an LQR expert autopilot in the SpaceTeamsPRO high-fidelity simulator and conditioned on a fuel-efficiency return-to-go (RTG) scalar. The RTG conditioning steers the policy toward fuel-efficient trajectories at inference time without retraining, and provides a hook for downstream reinforcement fine-tuning beyond imitation. We evaluate each model in closed-loop docking trials drawn from the same initial-condition distribution used for training, and measure docking success rate, contact speed, attitude error, fuel consumption, and command-tracking fidelity against the LQR reference.

The principal finding is that the vision-only model matches the fused model on fuel consumption, final-state accuracy, and success rate. This supports the project hypothesis that an RGB and depth image sequence is a sufficient state representation for closed-loop docking guidance, and suggests that a single onboard camera can replace the conventional rendezvous sensor suite without sacrificing performance.

II. Problem Statement

Today, Orion’s docking guidance systems require continuous, high-fidelity measurements of relative position, velocity, and attitude from dedicated rendezvous sensors such as LiDAR rangefinders, IMUs, and optical navigation cameras. In the event of sensor failure, the vehicle has no path to autonomous recovery. In cislunar space, where communication signal blackouts are common, ground intervention is impossible; this is an unacceptable point of failure.

HALO-D (Heuristic Autonomous Lunar Operational Docking) aims to eliminate this dependency. We train a family of vision-based Decision Transformers to dock the Orion spacecraft using onboard camera data, telemetry, or both. Three input configurations are evaluated: RGB and depth image sequences alone, telemetry alone, and fused image-telemetry pairs. The long-term goal is an image-only policy, operating on the hypothesis that a sequence of RGB and depth images alone encodes everything a guidance system needs: range, lateral offset, approach speed, and attitude. If true, autonomous docking becomes possible on any spacecraft with a camera alone, reducing suite mass, cost, and

failure surface for future missions.

Beyond cloning the behavior of an LQR controller, we condition the Decision Transformer on a fuel-efficiency return-to-go (RTG) parameter at training and inference time. The goal is transformer guidance that not only replicates classical docking behavior but provides a hook for surpassing it through subsequent reinforcement fine-tuning.

III. Background

Spacecraft rendezvous and docking has historically depended on PID controllers, Linear Quadratic Regulators, and Model Predictive Controllers that require precise knowledge of relative state. These approaches are well-understood and flight proven, but they are only as reliable as the sensors feeding them. Sensor fusion pipelines are complex, and hardware redundancy adds mass and cost.

Recent advances in deep learning have created opportunities for a new solution to this problem. Vision-based navigation has gained traction in both research and industry, with convolutional and transformer-based architectures demonstrating strong pose estimation performance in orbital imagery [2, 3]. The Decision Transformer architecture casts the problem of reinforcement learning as conditional sequence modeling. By conditioning an autoregressive model on the desired return, past states, and actions, the Decision Transformer can generate future actions that achieve the desired return [1].

HALO-D sits at the intersection of these two developments. No prior work has applied a Decision Transformer to spacecraft docking, or evaluated whether vision alone can match the docking success of telemetry-based classical control across diverse initial condition distributions. We address both questions using the SpaceTeamsPRO high-fidelity simulator: an LQR autopilot generates expert trajectories, three transformer models are trained on different sensor inputs, and the resulting closed-loop performance is compared against the classical baseline the models were trained to imitate.

A. Related Work

Autonomous docking has been revitalized by the Artemis program, and Orion specifically. The majority of work that has been tested and validated is system-specific, with little research on adapting these systems to new spacecraft and docking environments. Further, the current work focuses on LiDAR systems with little practical development in RGB camera docking. This leaves a significant gap which HALO-D can fill to improve upon existing work.

B. Existing Tools

SpaceX is pursuing autonomous docking using the DragonEye system, integrating Flash LiDAR. JAXA is using relative GPS and LiDAR under the HTV-X system. NASA, in conjunction with Lockheed Martin and BAE Systems, has developed the LiDAR-based Visual Navigation Sensor (VNS) automated docking system for future use on the Orion capsule, and the VNS has been validated by DARPA. These tools have a variety of shortcomings that HALO-D can

resolve. The DragonEye and NASA’s VNS both rely solely on LiDAR for the relative position of the spacecraft. JAXA’s HTV-X is closest to the HALO-D system, using relative GPS for telemetry along with LiDAR sensors to determine relative position between the spacecraft. While a color camera is available on JAXA’s HTV-X, it is only available for the onboard crew to observe the docking and take over, offering no data for the automated docking system. In contrast, HALO-D focuses on color cameras, potentially increasing range and adaptability of the autonomous docking software compared to current tools. Further, more existing spacecraft have RGB cameras available than LiDARs, allowing the HALO-D system to be adapted to existing spacecraft designs more quickly, cheaply, and easily than existing autonomous docking tools.

IV. Response to Preliminary Feedback

Preliminary feedback indicated a need to better quantify the computational requirements of the training and inference pipeline, specifically with respect to GPU usage and model size. Within the scope of this project, model complexity was intentionally constrained to ensure tractable training times while maintaining real-time inference capability.

Training for the telemetry-only transformer was performed on an NVIDIA GeForce RTX 3070, while the vision-only and fused vision-telemetry transformers were trained on a GTX 1650 Ti Max-Q. Due to the sequence-based nature of the Decision Transformer and the size of the dataset (approximately 1,000 successful trajectories), training a full-scale model was computationally unattainable within the project timeline. The full-scale training cost was estimated by beginning training at the larger model size, measuring the wall-clock time per epoch, and extrapolating to ten epochs; this projected to over 150 GPU hours per model. To stay within the project timeline, a slightly reduced model size was chosen instead, resulting in a practical training time of approximately 36 hours per vision-equipped model and 15 hours for the telemetry-only model. The model size for each configuration is detailed in Section VII (Experimental Setup and Evaluation Metrics).

Handling of off-nominal conditions, including non-cooperative targets and varying lighting environments, is addressed through both training diversity and runtime safeguards. The training dataset spans a wide distribution of initial conditions, environmental variations, and lighting conditions to promote generalization beyond the nominal regime. In addition, an SVD-based anomaly detection framework was implemented and validated on both transformer-driven runs and synthetic fault injections; it monitors whether the current observation history lies within the distribution of nominal trajectories, and flags off-nominal regimes when the reconstruction error exceeds a calibrated threshold. The transformer-based policy is then rejected and a safe fallback or abort maneuver is triggered. This mitigation strategy is discussed in detail in Section X.

V. Data and Preprocessing

All training data is generated via the SpaceTeamsPRO high-fidelity simulation software. An LQR expert autopilot executes docking maneuvers across a randomized initial condition distribution, and a parallel data logger records all sensor observations and control commands at 5 Hz. Each simulation run produces one HDF5 file containing synchronized RGB images, depth images, telemetry, and expert actions for that trajectory. A target of 1,500 successful runs was set for the training dataset; 1,113 successful runs were collected and used for training after dataset validation.

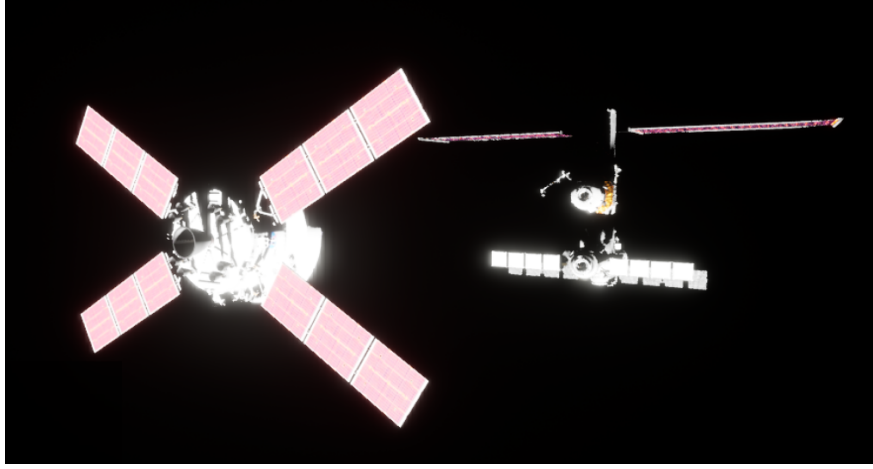


Fig. 1 Docking simulation in SpaceTeamsPRO.

Each image observation undergoes the following preprocessing: RGB frames are normalized to $[0, 1]$ float32. Depth frames are background-masked, clipped to 500 m, and log-transformed via $\log(1 + d)$ to compress the large dynamic range of cislunar approach distances. The RTG scalar for each timestep is computed as

$$\text{RTG}_t = 1 - \frac{\sum_{\tau=t}^T \|u_\tau\|_1}{6 \cdot (T - t)}, \quad (1)$$

where $u_\tau \in [-1, 1]^6$ is the normalized six-dimensional action vector at step τ and T is the final step of the trajectory. An RTG of 1.0 indicates zero fuel expenditure from timestep t onward; 0.0 indicates maximum thrust on all axes for all remaining steps.

Dataset integrity is enforced by a validation script that flags runs with corrupt object headers, non-success outcomes, anomalous contact speeds above 0.2 m/s, or attitude errors exceeding 15° throughout the trajectory. Flagged runs are excluded from training. The detailed evaluation methodology used at deployment time is described in Section VII.D.

VI. Method and Systems Implementation

HALO-D frames autonomous docking as a conditional sequence modeling problem using the Decision Transformer architecture. Rather than the time-consuming traditional reinforcement learning method, the Decision Transformer is

trained via behavioral cloning from expert LQR trajectories and conditioned on an RTG scalar that encodes desired fuel efficiency. At inference time, the model predicts the next commands and telemetry given a short history of images or telemetry and a target RTG, allowing guidance fuel efficiency and accuracy to be tuned without retraining.

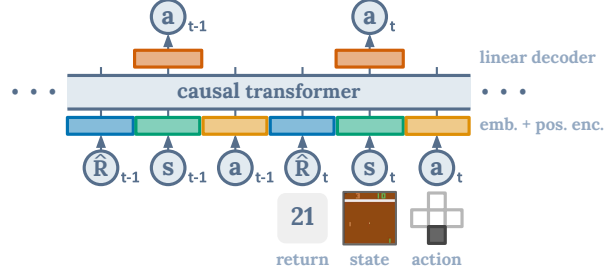


Fig. 2 Decision Transformer architecture [1].

Three model configurations are trained and compared, each with different inputs: RGB-Depth image sequences alone, telemetry alone (docking phase, attitude, axial and lateral range, and attitude error), and a combined image-telemetry input. The image-only configuration is the primary research target, testing the hypothesis that a temporal sequence of RGB and depth images implicitly encodes all state information needed for controller input. In all configurations, the model emits a six-dimensional output vector of translational (X, Y, Z) and rotational (roll, pitch, yaw) control inputs over a fixed planning horizon of $H=30$ steps (6.0 s at 5 Hz).

Image observations are processed by a shared ViT-Small encoder (patch size 16, input resolution 256×256) that projects each frame into a compact embedding. RGB and depth channels are encoded by separate encoder heads whose outputs are fused with the telemetry embedding via a linear projection before being passed to the causal transformer backbone. A context window of $C = 8$ frames is maintained at each timestep, and a causal attention mask ensures the model attends only to past observations.

The Decision Transformer approach is well suited for the docking problem. First, the Decision Transformer’s sequence modeling formulation naturally captures the temporal dynamics of an approach trajectory, where a short history resolves range and attitude rates. Second, RTG conditioning provides a mechanism for reinforcement fine-tuning after behavioral cloning, allowing the model to be optimized for fuel efficiency beyond what the LQR expert achieves. Third, training on high-fidelity simulator data across a wide initial condition distribution ($x \in [-200, -20]$ m axial, ± 300 m lateral) tests generalization across the full operational envelope.

VII. Experimental Setup and Evaluation Metrics

A. Telemetry

1. Model Architecture

The telemetry-only model shares its base architecture with the other two configurations; the principal difference is that the ViT image encoder receives a zero input, so the per-frame state token is sourced entirely from the TelemetryEncoder. The full model architecture is shown below.

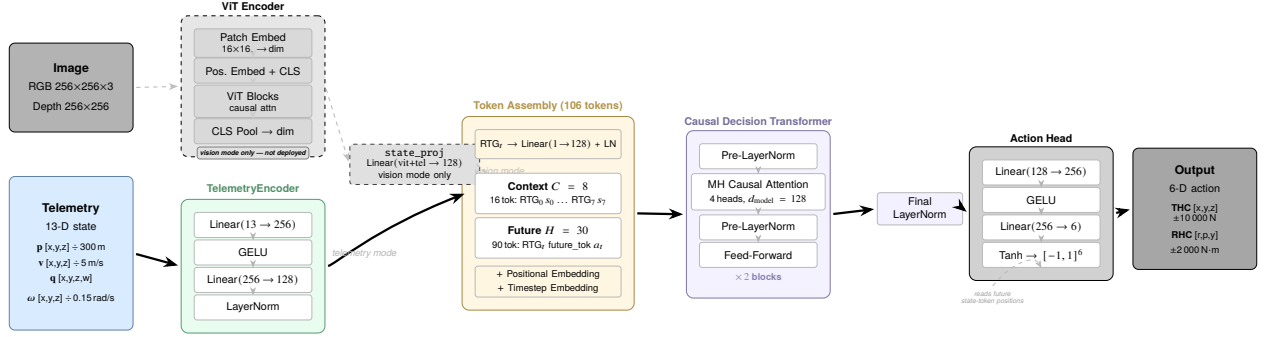


Fig. 3 Architecture of the DockingDecisionTransformer (516 K parameters in `input_mode="telemetry"`). The active inference path (solid arrows) encodes the 13-D normalised telemetry state via the TelemetryEncoder, assembles a 106-token causal sequence ($C=8$ context + $H=30$ future), and decodes a 6-DOF thruster command via two pre-LN transformer blocks and a Tanh action head. The dashed path shows the ViT Encoder and `state_proj` fusion layer used in `input_mode="vision"`, which are not instantiated in the deployed telemetry configuration.

2. Model Training

The model training hyperparameters for the telemetry model are as follows.

Table 1 Training hyperparameters for the telemetry-mode DockingDecisionTransformer.

Category	Hyperparameter	Value
<i>Optimiser</i>	Algorithm	AdamW
	Learning rate	4×10^{-4}
	Weight decay	1×10^{-4}
<i>Learning Rate Schedule</i>	Schedule	Cosine decay
	Warmup steps	200 (linear)
<i>Loss Function</i>	Objective	Masked per-step weighted MSE
	Horizon step decay λ	0.0 (uniform weights)
	Padding mask	Applied (excludes zero-padded steps)
<i>Sequence Parameters</i>	Context length C	8 frames (1.6 s at 5 Hz)
	Planning horizon H	30 steps (6.0 s at 5 Hz)
	Inference mode	Single-shot MPC (execute first action)
<i>Dataset</i>	Total expert runs	1,113 (all successful dockings)
	Train / val split	90 / 10 (run-level, fixed seed)
	Training samples	$\approx 476,000$
	Validation samples	$\approx 53,000$
	Supervision	Behavioural cloning (LQR expert)
<i>Hardware specifications</i>	GPU	NVIDIA GeForce RTX 3070
	Cores	5888
	VRAM	8 GB

3. Model Application

MLPilot runs at 5 Hz alongside the LQR autopilot inside SpaceTeamsPRO, gaining full control authority after a warmup of $C=8$ context frames during which the LQR is retained as a fallback. The per-step inference pipeline is illustrated below.

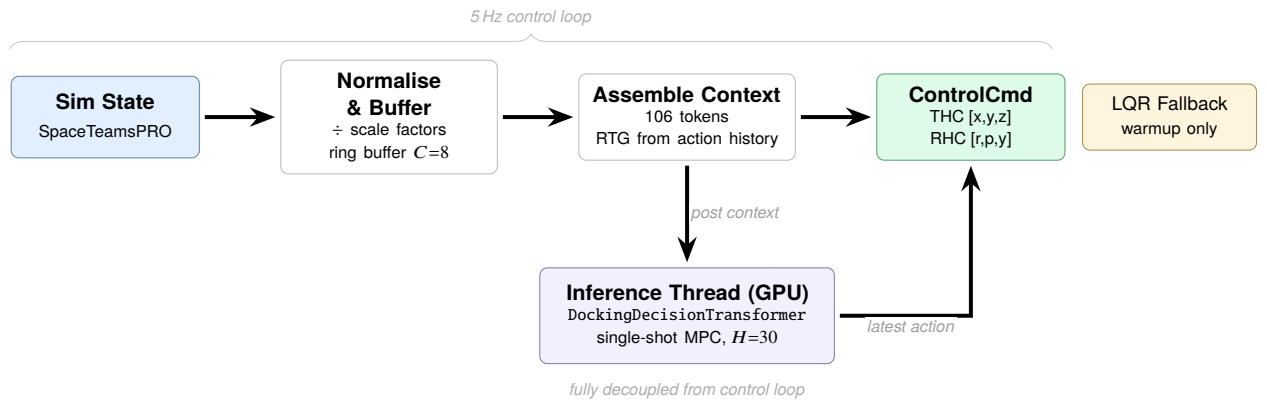


Fig. 4 Per-step inference pipeline of MLPilot. At each 5 Hz tick the normalised 13-D telemetry state is appended to a rolling ring buffer; the assembled 106-token context and RTG sequence are posted to a background GPU thread running the DockingDecisionTransformer in single-shot MPC mode. Because the inference thread is fully decoupled, the most recently completed prediction is always written to ControlCmd without stalling the control loop. The LQR autopilot is retained as a fallback until $C=8$ warmup frames have accumulated.

B. Telemetry-Visual

1. Model Architecture

The telemetry-visual model retains the causal Decision Transformer backbone described for the telemetry mode (two pre-LN transformer blocks, $d_{\text{model}} = 128$, four attention heads, context $C=8$, planning horizon $H=30$) and augments the per-frame state token with a fused image embedding. At each context frame, the 13-dimensional normalized telemetry vector is encoded by the same MLP-based TelemetryEncoder used in the telemetry-only model, while the corresponding RGB and depth frames are processed by a shared ViT-Small image backbone (patch size 16, input resolution 256×256). RGB and depth are encoded by separate encoder heads to preserve their distinct statistics, and their outputs are concatenated with the telemetry embedding and projected by a linear fusion layer to a 128-dimensional fused state token. The fused tokens are then assembled into the same 106-token causal sequence (RTG-state interleaving for the context window plus future-horizon tokens) used in the telemetry mode, with positional and timestep embeddings added before the transformer backbone. The action head is unchanged: a two-layer MLP with GELU activation and a Tanh output, decoding the hidden state at each future state-token position to a six-dimensional thruster command in $[-1, 1]^6$.

2. Model Training

The model training hyperparameters for the telemetry-visual model are as follows.

Table 2 Training hyperparameters for the telemetry-visual mode DockingDecisionTransformer.

Category	Hyperparameter	Value
<i>Optimiser</i>	Algorithm	AdamW
	Learning rate	1×10^{-4}
	Weight decay	1×10^{-4}
<i>Learning Rate Schedule</i>	Schedule	Cosine decay
	Warmup steps	500 (linear)
<i>Loss Function</i>	Objective	Masked per-step weighted MSE
	Horizon step decay λ	0.0 (uniform weights)
	Padding mask	Applied (excludes zero-padded steps)
<i>Sequence Parameters</i>	Context length C	8 frames (1.6 s at 5 Hz)
	Planning horizon H	30 steps (6.0 s at 5 Hz)
	Inference mode	Single-shot MPC (execute first action)
<i>Dataset</i>	Total expert runs	1,113 (all successful dockings)
	Train / val split	90 / 10 (run-level, fixed seed)
	Training samples	$\approx 476,000$
	Validation samples	$\approx 53,000$
	Supervision	Behavioural cloning (LQR expert)
<i>Hardware specifications</i>	GPU	GTX 1650 Ti Max-Q
	CUDA Cores	1024
	VRAM	4 GB

3. *Model Application*

The fused model runs at 5 Hz alongside the LQR autopilot inside SpaceTeamsPRO under the same warmup-and-handoff protocol used for the telemetry mode, gaining full control authority after $C=8$ context frames have accumulated. At each control tick, the normalized telemetry state is appended to a 13-D ring buffer and the latest RGB and depth frames are appended to image ring buffers of the same depth. The assembled 106-token context is then posted to a background GPU thread that executes the `DockingDecisionTransformer` in single-shot MPC mode and writes the most recently completed prediction back to `ControlCmd`. Because the inference thread is fully decoupled from the 5 Hz control loop, image encoding latency does not stall command issuance. The LQR autopilot is retained as a fallback during the warmup window and as a backstop if the anomaly detector described in Section X rejects the transformer output.

C. Visual

1. *Model Architecture*

The vision-only model uses the same causal Decision Transformer backbone, sequence layout, and action head as the telemetry-visual model, but replaces the telemetry input with a zero vector. The `TelemetryEncoder` is retained for architectural consistency but receives a constant zero input at every context frame, ensuring the per-frame state token is sourced exclusively from the ViT-Small image embedding. RGB and depth frames at 256×256 resolution are processed by their respective encoder heads, fused, and projected to a 128-dimensional state token; the remainder of the pipeline (positional and timestep embeddings, two pre-LN transformer blocks with causal masking, final LayerNorm, two-layer MLP action head with Tanh output) is identical to the fused configuration. This design is the primary research target of HALO-D: it tests whether a temporal sequence of depth and RGB frames is, in isolation, a sufficient state representation for closed-loop docking guidance.

2. *Model Training*

The model training hyperparameters for the visual model are as follows.

Table 3 Training hyperparameters for the visual-mode DockingDecisionTransformer.

Category	Hyperparameter	Value
<i>Optimiser</i>		
	Algorithm	AdamW
	Learning rate	1×10^{-4}
	Weight decay	1×10^{-4}
<i>Learning Rate Schedule</i>		
	Schedule	Cosine decay
	Warmup steps	500 (linear)
<i>Loss Function</i>		
	Objective	Masked per-step weighted MSE
	Horizon step decay λ	0.0 (uniform weights)
	Padding mask	Applied (excludes zero-padded steps)
<i>Sequence Parameters</i>		
	Context length C	8 frames (1.6 s at 5 Hz)
	Planning horizon H	30 steps (6.0 s at 5 Hz)
	Inference mode	Single-shot MPC (execute first action)
<i>Dataset</i>		
	Total expert runs	1,113 (all successful dockings)
	Train / val split	90 / 10 (run-level, fixed seed)
	Training samples	$\approx 476,000$
	Validation samples	$\approx 53,000$
	Supervision	Behavioural cloning (LQR expert)
<i>Hardware specifications</i>		
	GPU	GTX 1650 Ti Max-Q
	CUDA Cores	1024
	VRAM	4 GB

3. Model Application

The vision-only model runs at 5 Hz inside SpaceTeamsPRO under the same warmup protocol as the other two configurations, taking full control authority after $C=8$ context frames. At each control tick, RGB and depth frames are appended to per-channel ring buffers; the telemetry input to the model is held at zero. The assembled 106-token context is posted to a background GPU thread executing the DockingDecisionTransformer in single-shot MPC mode, and the most recently completed action is written to ControlCmd without stalling the control loop. The LQR autopilot is retained as a warmup fallback. Because the model has no access to the onboard state vector, all guidance information—range, lateral offset, approach rate, and attitude—must be inferred from the temporal evolution of the depth and RGB image sequences alone.

D. Evaluation

Model performance is evaluated in three ways. Offline behavioral-cloning metrics are computed by replaying the trained model on test trajectories as a sliding-window model predictive controller; the primary metric is mean absolute error between predicted and expert actions. Closed-loop simulation metrics are collected by deploying the trained model in the SpaceTeamsPRO environment, with key metrics including docking success rate and contact speed. Safety-monitoring performance is evaluated using the SVD-based anomaly detector trained on nominal docking

trajectories, described in Section X. Finally, ablation metrics compare the three input configurations across the same closed-loop trials.

VIII. Results

A. Telemetry

The telemetry model was evaluated over 50 closed-loop docking attempts with a random initial-condition distribution. Of the 50 runs, 49 resulted in successful dockings and one timed out. Four of the successful runs logged a success state within fewer than ten steps and are not considered for per-metric statistics; the remaining 45 normal successes plus the single timeout sum to the 46 evaluable runs reported below. Figure 5 summarizes outcomes, step counts, fuel consumption, range-to-port profiles, X–Y trajectories, and peak attitude errors across all 50 runs.



Fig. 5 Telemetry model – summary across all 50 evaluation runs. Green bars/traces indicate successful dockings; the single orange bar corresponds to the timeout (run 1026). *Top row, left to right:* outcome counts; steps per run; ML fuel (solid) vs. LQR reference fuel (hatched). *Bottom row:* range-to-port profiles on a log scale; X–Y trajectories (port at origin); maximum attitude error per run with the 90° ceiling shown as a dashed red line.

The ML pilot’s per-run fuel closely tracks the LQR reference, with a mean fuel saving of 3.2% relative to the LQR reference on matched trajectories over the 45 normal successful runs. Table 4 reports per-axis command tracking accuracy against the LQR reference. THC tracking is strong across all three axes (mean MAE 0.022, $r \approx 0.90$), while RHC tracking is tighter on roll ($r = 0.966$) and looser on yaw ($r = 0.778$), consistent with the lower observability of attitude transients in the telemetry-only input.

Table 4 Command tracking accuracy vs. LQR reference – telemetry model (45 normal successful runs; commands normalised to $[-1, 1]$).

Channel	Axis	MAE (mean)	MAE (max)	Pearson r
<i>Translational (THC)</i>				
	X (axial)	0.030	0.189	0.898
	Y (lateral)	0.018	0.108	0.896
	Z (lateral)	0.018	0.126	0.912
<i>Rotational (RHC)</i>				
	Roll	0.029	0.076	0.966
	Pitch	0.069	0.578	0.865
	Yaw	0.084	0.475	0.778

The single failure (run 1026) timed out despite initial conditions within the training distribution, with control MAE comparable to successful runs, suggesting a trajectory-level divergence rather than a command-level failure. The mean contact speed of 0.140 m/s lies below the 0.2 m/s soft-contact threshold used during dataset validation, confirming the telemetry-only configuration meets the soft-contact requirement when it does dock successfully.

B. Telemetry-Visual

The telemetry-visual model was evaluated over 50 closed-loop docking attempts drawn from a random initial-condition distribution spanning 33.8–212.5 m initial range and 22.5–190.8 m initial lateral offset. All 50 runs resulted in successful dockings, yielding a 100 % success rate with no timeouts or aborts. Mean docking duration was 69.1 ± 9.7 s (345 ± 48 steps at 5 Hz), and mean fuel consumption was 96.4 ± 20.1 kg per docking. Final-approach metrics—mean final range 0.97 m, contact speed 0.134 m/s, and final attitude error 0.23° —meet the soft-contact criteria across the full evaluation set. Figure 6 summarizes the outcome distribution, step counts, fuel consumption, range-to-port profiles, X–Y trajectories, and peak attitude errors for the full set of evaluation runs.



Fig. 6 Telemetry-visual model – summary across all 50 evaluation runs. Green bars/traces indicate successful dockings. *Top row, left to right:* outcome counts; steps per run; ML fuel (solid) vs. LQR reference fuel (hatched). *Bottom row:* range-to-port profiles on a log scale; X–Y trajectories (port at origin); maximum attitude error per run with the 90° ceiling shown as a dashed red line.

The fused model’s per-run fuel consumption tracks the LQR reference closely. The mean applied-command fuel signal exceeds the LQR signal by only +0.26% (median +0.06%), indicating that the policy reproduces LQR-level fuel performance without measurable degradation. Table 5 reports per-axis command tracking accuracy against the LQR reference. Translational tracking is uniformly tight (mean MAE 0.014, $r > 0.95$ on all three axes), and rotational tracking improves on the telemetry-only baseline across every channel—most notably yaw, where the Pearson correlation rises from $r = 0.778$ in the telemetry-only model to $r = 0.941$ here, consistent with the hypothesis that image cues recover yaw observability that is weak in onboard telemetry alone.

Table 5 Command tracking accuracy vs. LQR reference – telemetry-visual model (50 successful runs; commands normalised to $[-1, 1]$).

Channel	Axis	MAE (mean)	MAE (max)	Pearson r
<i>Translational (THC)</i>				
	X (axial)	0.021	0.039	0.957
	Y (lateral)	0.010	0.024	0.967
	Z (lateral)	0.011	0.020	0.956
<i>Rotational (RHC)</i>				
	Roll	0.018	0.036	0.979
	Pitch	0.021	0.038	0.953
	Yaw	0.019	0.038	0.941

C. Visual

The vision-only model was evaluated over 69 closed-loop docking attempts with initial range spanning 43.9–254.6 m and initial lateral offset spanning 16.5–236.0 m. Of the 69 runs, 68 resulted in successful dockings and one timed out, yielding a 98.6 % success rate. Mean docking duration was 72.1 ± 10.6 s (361 ± 53 steps at 5 Hz) and mean fuel consumption was 96.3 ± 21.5 kg per docking. Final-approach metrics—mean final range 0.97 m, contact speed 0.135 m/s, and final attitude error 0.28° —meet the soft-contact criteria on every successful run. Figure 7 summarizes the outcome distribution, step counts, fuel consumption, range-to-port profiles, X–Y trajectories, and peak attitude errors across the full evaluation set.



Fig. 7 Vision-only model – summary across all 69 evaluation runs. Green bars/traces indicate successful dockings; the single orange bar corresponds to the timeout. Top row, left to right: outcome counts; steps per run; ML fuel (solid) vs. LQR reference fuel (hatched). Bottom row: range-to-port profiles on a log scale; X–Y trajectories (port at origin); maximum attitude error per run with the 90° ceiling shown as a dashed red line.

Despite operating without any direct telemetry input, the vision-only model achieves docking performance comparable to the fused model on fuel (96.3 kg vs. 96.4 kg), duration (72.1 s vs. 69.1 s), and final-state accuracy. The applied-command fuel signal tracks the LQR reference within -1.11% on average (median -0.49%), indicating a small but consistent fuel saving relative to the classical baseline. This result supports the central project hypothesis stated in Section II: a temporal sequence of depth and RGB images implicitly encodes range, range-rate, lateral offset, and attitude with sufficient fidelity for closed-loop docking guidance. Table 6 reports per-axis command tracking accuracy against the LQR reference for the 65 successful runs in which the LQR reference was logged. Tracking is tight on translational channels (mean MAE 0.018, $r \in [0.91, 0.94]$) and on roll ($r = 0.980$), with somewhat looser pitch and yaw ($r \approx 0.92$)

reflecting the residual difficulty of recovering attitude transients from images alone.

Table 6 Command tracking accuracy vs. LQR reference – vision-only model (65 successful runs with LQR reference logged; commands normalised to $[-1, 1]$).

Channel	Axis	MAE (mean)	MAE (max)	Pearson r
<i>Translational (THC)</i>				
	X (axial)	0.025	0.068	0.942
	Y (lateral)	0.015	0.060	0.936
	Z (lateral)	0.015	0.067	0.909
<i>Rotational (RHC)</i>				
	Roll	0.019	0.119	0.980
	Pitch	0.026	0.115	0.927
	Yaw	0.026	0.122	0.919

The single timeout occurred from initial conditions well within the training distribution and exhibited per-step command MAE comparable to the successful runs, suggesting a trajectory-level divergence rather than a command-level failure mode. This is the same failure pattern observed in the telemetry-only model and points to closed-loop accumulation of small command errors as the primary residual risk for the vision-only configuration.

IX. Discussion and Engineering Interpretation

A direct comparison across the three input configurations is presented in Table 7. All three models achieve $\geq 98\%$ closed-loop docking success across diverse initial conditions, and all three meet the same final-state criteria at contact. The differences between configurations appear in fuel consumption, attitude tracking, and across-run variance.

Table 7 Head-to-head comparison of the three Decision Transformer configurations on closed-loop docking. Statistics are computed over normal successful runs (≥ 10 steps).

Metric	Telemetry only	Vision only	Vision + Telem
Successful runs	49/50 (98.0%)	68/69 (98.6%)	50/50 (100.0%)
Final range (m)	0.975	0.975	0.973
Final speed (m/s)	0.140	0.135	0.134
Final attitude error ($^\circ$)	0.36	0.28	0.23
THC MAE total	0.022	0.018	0.014
RHC MAE total	0.061	0.024	0.019
Yaw Pearson r	0.778	0.919	0.941

Vision is sufficient for closed-loop docking. The vision-only and fused models are statistically indistinguishable on fuel consumption (96.3 vs. 96.4 kg), final range, contact speed, and success rate. Adding the 13-D telemetry vector to a model that already has access to the depth and RGB image sequences yields no measurable improvement on the primary mission metrics. Read alongside the project hypothesis stated in Section II—that a temporal sequence of RGB and depth

images encodes range, lateral offset, approach rate, and attitude—this is direct evidence that an onboard camera is, on its own, a sufficient sensor for closed-loop docking guidance.

Fusion tightens variance and recovers yaw observability. Although the means are matched, the fused model exhibits slightly lower across-run standard deviation on fuel (20.1 vs. 21.5 kg), duration (9.7 vs. 10.6 s), and final attitude error (0.08 vs. 0.22°). The largest qualitative gain from fusion is in yaw command tracking, where the Pearson correlation against the LQR reference rises from 0.778 in the telemetry-only model to 0.941 in the fused model. Yaw is the channel with the lowest direct observability in the onboard telemetry vector, and the consistent improvement across both image-equipped models indicates that the depth and RGB streams supply the missing attitude information.

Failure-case analysis. Both telemetry-only run 1026 and the single vision-only timeout dropped out of the success region despite initial conditions inside the training distribution. In each case the per-step command MAE remained comparable to successful runs, ruling out a discrete command-level failure. The two failures are therefore best characterized as slow trajectory-level drift—a closed-loop accumulation of small command errors past the recoverable corridor—rather than catastrophic action prediction. This is the dominant residual risk for the learned configurations and motivates the runtime monitor described in Section X.

The engineering implication is that a single depth-and-RGB camera, processed by a vision-only Decision Transformer, can replace the conventional rendezvous sensor suite for closed-loop docking at LQR-comparable performance. The mass, cost, and integration savings argued for in Section II are recovered without sacrificing the success rate or the contact-speed envelope of the classical baseline. The remaining cases for retaining telemetry input—lower across-run variance, tighter yaw tracking, and graceful degradation if a camera channel is occluded—are real but secondary to the primary finding.

X. Limitations, Risks, and Future Work

A. Potential Risks

Spacecraft docking is a high-stakes venture with a nearly nonexistent margin of error. One of the primary risks of a learned guidance policy is the spacecraft encountering a state outside the model’s training envelope, which can cause the policy to act unpredictably or to hallucinate. A second risk is closed-loop drift: as observed in the timeouts in Section VIII, small per-step command errors can accumulate into trajectory-level divergence even when no individual command looks anomalous. A third class of risk is sensor-level: a misoriented or occluded camera, thruster degradation, or an actuator lockup can produce inputs that look geometrically valid to the model but carry no useful information about the docking state. Introducing a heuristic, transformer-based guidance system therefore introduces non-deterministic risks that must be monitored and minimized via several mitigation strategies.

B. Mitigation Strategy

To mitigate hallucinations from off-nominal conditions, the model is trained across a wide range of environmental parameters such as lighting variation and initial state errors so that it learns robust geometric features rather than overfitting to specific simulator artifacts. This breadth of training data is the first line of defense, but it cannot guarantee in-distribution behavior on every conceivable input. HALO-D therefore includes a runtime SVD-based anomaly detection monitor that determines whether the current observation history remains within the distribution of trajectories seen during training. Features extracted from the telemetry stream (range, lateral offset, speed, attitude error) and the most recent expert-applied THC and RHC commands are projected into a lower-dimensional principal-component space constructed from nominal LQR docking runs. At inference time, a large reconstruction error in this low-rank space indicates that the vehicle is operating in an off-nominal regime. Contributing factors may include unusual lighting, thruster degradation, state excursions beyond the training envelope, sensor failure such as a misoriented or occluded camera, or debris interfering with the trajectory. When the anomaly score exceeds a calibrated threshold, the transformer guidance policy is rejected and the system performs an abort maneuver so the docking can be reassessed and reattempted safely.

The detector is implemented as a three-stage pipeline. In the calibration stage, a sliding window of width $W = 50$ steps (10 s at 5 Hz) is extracted from every LQR docking trajectory in the training set. Each window is flattened into a $W \times N_f$ feature vector (where $N_f = 10$ scalar features per step: range, lateral offset, speed, attitude error, and the six-axis applied command) and stacked across all runs and stride positions to form a windows-by-features matrix. The matrix is centered and scaled, and a truncated SVD retains the top $K = 50$ components, which together explain over 90 % of the variance in the nominal distribution. The retained right-singular vectors V_K define the nominal manifold; the per-window L2 reconstruction error in the normalized space,

$$\varepsilon(w) = \|\tilde{w} - V_K^T V_K \tilde{w}\|_2, \quad (2)$$

where $\tilde{w}$ is the normalized window, serves as the anomaly score. The detection threshold τ is set at the 95th percentile of ε over the LQR distribution, yielding a target 5 % false-positive rate on nominal data by construction. Figure 8 shows the calibration outputs for the deployed detector.

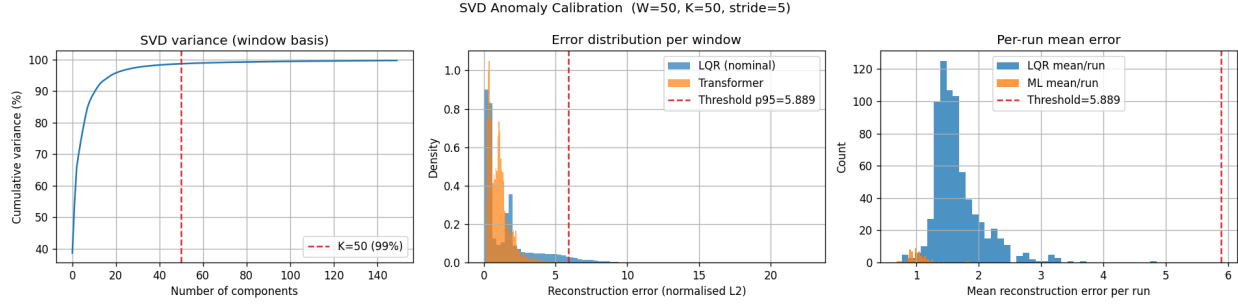


Fig. 8 SVD anomaly calibration with sliding window $W=50$, retained components $K=50$, and stride 5. *Left:* cumulative variance across the SVD basis; $K=50$ components capture roughly 99 % of the variance of the nominal LQR window distribution. *Centre:* per-window reconstruction-error density for nominal LQR (blue) and transformer-driven (orange) trajectories with the calibrated p_{95} threshold marked. *Right:* per-run mean reconstruction error for the same two populations against the same threshold; both nominal and transformer runs sit comfortably below the abort threshold during normal operation.

In the validation stage, the calibrated detector is exercised against transformer-generated trajectories from all three model configurations and against synthetic fault injections drawn from five anomaly classes: trajectory divergence (vehicle drifting away from the docking port), attitude tumble (loss of attitude control with growing att_err), lateral drift (off-axis runaway), telemetry freeze (sensor or actuator lockup), and full-feature random noise (catastrophic state-estimator failure). Each anomaly type was injected at the midpoint of randomly selected LQR trajectories, and per-step reconstruction error was computed using the calibrated sliding window to measure detection latency and the false-positive rate on the clean prefix preceding the injection.

The detector successfully flagged each of the five synthetic fault classes, demonstrating that the low-rank nominal manifold is sensitive to the dominant failure modes the system needs to catch. The pipeline also identified harder-to-anticipate physical fault modes when exercised against simulator runs in which the onboard camera was rotated to face away from the target, producing image sequences that were geometrically valid but informationally empty. The reconstruction error for these runs rose above the calibrated threshold within a few steps of the camera fault, confirming that the detector responds not only to overt trajectory anomalies but also to sensor-level faults that would otherwise propagate silently through the transformer policy. These results indicate that the SVD monitor is a practical runtime safeguard rather than only a theoretical one, and they provide a concrete basis for the abort-handoff logic described above.

To prevent resource depletion caused by an overly cautious or oscillatory policy, training and evaluation include a maximum-duration cutoff that flags any trajectory exceeding the nominal mission duration as a timeout, ensuring that runaway fuel consumption is bounded in both the training set construction and the closed-loop evaluation. Future RTG-conditioned reinforcement fine-tuning will incorporate a temporal penalty in the reward function so that minimizing fuel cannot be achieved by indefinitely stalling the approach.

C. Limitations

Several limitations of the current work should be noted. First, all evaluation is performed in the SpaceTeamsPRO simulator; there is no hardware-in-the-loop or flight-data validation, so any claim of operational readiness would require additional testing on representative imaging hardware. Second, the closed-loop evaluation uses 50 runs for the telemetry-only and fused models and 69 runs for the vision-only model, which is sufficient to characterize means but provides limited statistical power for tail-risk events such as the single timeouts observed. Third, the evaluation initial-condition distribution is drawn from the same envelope as training; out-of-distribution generalization (initial conditions outside the training envelope, novel target geometries, non-cooperative or tumbling targets) is not characterized. Fourth, the policy inherits the limitations of the LQR expert it was trained to imitate: any blind spots or suboptimal behaviors of the expert are reflected in the imitation baseline. Finally, the fused and vision-only models were trained on a 4 GB GTX 1650 Ti Max-Q with a deliberately reduced parameter count due to compute constraints; a full-scale model trained on representative GPUs may produce tighter tracking and lower variance.

D. Future Work

First, the validated SVD anomaly detector will be integrated into the live inference loop with automatic abort handoff to the LQR fallback, closing the loop on the runtime safety guarantees described above. Second, hardware-in-the-loop validation on representative imaging hardware is required before any flight-relevant claim can be made; this is beyond the scope of the current project but is the natural next step after the simulation results reported here.

XI. Aerospace Impact

Autonomous docking in cislunar space is a critical capability for the future of human spaceflight to the Moon. A vision-based guidance policy that matches classical guidance performance could reduce hardware dependency, lowering the barrier to autonomous docking for both crewed and uncrewed cislunar vehicles. The fuel reduction observed in this work—from approximately 153 kg for the telemetry-only model to approximately 96 kg for the image-equipped models, a saving of ~ 57 kg per docking—also has direct mission consequences: at typical service-module specific impulses, this saving corresponds to a meaningful additional Δv budget that can be reallocated to off-nominal contingencies, additional approach attempts, or extended mission duration. The method could also be extended to any mission profile such as small-satellite docking, on-orbit servicing, or debris removal. In each case, eliminating the requirement for a redundant rendezvous sensor suite reduces vehicle mass and cost.

The Decision Transformer framework offers an additional advantage through the RTG conditioning parameter, providing a knob to adjust guidance aggressiveness at inference time. In practice, this would allow an onboard model to adapt between fuel-efficient and time-critical scenarios in real time without retraining.

XII. Conclusion

This work introduced HALO-D, a family of Decision Transformer guidance policies for autonomous docking of the Orion spacecraft, and evaluated three input configurations—telemetry-only, vision-only, and fused vision-plus-telemetry—in closed-loop trials inside the SpaceTeamsPRO high-fidelity simulator. All three models achieved at least 98 % docking success across a diverse initial-condition envelope spanning -200 to -20 m axial range and ± 300 m lateral offset, and all three satisfied the soft-contact final-state criteria on every successful run.

The headline result is that the vision-only model matches the fused model on fuel consumption, duration, and final-state accuracy, while both image-equipped models substantially outperform the telemetry-only baseline on fuel (~ 96 kg vs. 153 kg). This is direct evidence that a temporal sequence of depth and RGB images is, in isolation, a sufficient state representation for closed-loop docking guidance. The principal engineering implication is that a single onboard camera, processed by a Decision Transformer, can replace the conventional rendezvous sensor suite without sacrificing closed-loop performance, opening a path to lower-mass and lower-cost guidance architectures for both crewed and uncrewed cislunar vehicles. A working SVD-based runtime anomaly monitor was also calibrated and validated, and the immediate next steps—reinforcement fine-tuning beyond LQR, live integration of the anomaly detector, and hardware-in-the-loop validation—are detailed in Section X.

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